

second touch plate. R1 act as a feedback resistor along with the skin when the touch plates make contact with the skin. So the gain of IC1 depends on the feedback provided by R1 and the skin.

In the inverting mode of IC1, a positive input voltage to its pin 2 through the feedback network makes its output low. If the skin offers very high resistance in the relaxed state, input voltage to pin 2 reduces and the output remains high. Thus the gain of IC1 varies depending on the current passing through the skin, which, in turn, depends on the skin response and emotional state.

In the standby state, touch plates are free. As there is no feedback to IC1, it gives a high output (around 6 volts), which is indicated by shifting of the meter to right side.

When the touch plates are shorted by the skin, the feedback circuit completes and the output voltage reduces to 4 volts or less depending on the resistance of the skin. Since the feedback network has a fixed resistor (R1) and VR1 is set to a fixed resistance value,

the current flowing through it depends only on the resistance of the skin. The output from IC1 is displayed on a sensitive moving coil meter (VU meter). By varying preset VR2, you can adjust the sensitivity of the meter.

For easy visual observation, an LED display is also included. IC LM3915 (IC2) is used to give a logarithmic display through LED indications. It can sink current from pin 18 to pin 10 with each increment of 125 millivolts at its input pin 5. Using VR3 you can adjust the input voltage of IC2, while using VR4 you can control the brightness of the LEDs.

In practice, the circuit provides both meter reading and LED indications. If the LED display is not needed, IC2 can be omitted.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet with touch pads glued on the top, 5-10 mm apart. Touch pads can be any type of conducting plates, such as aluminium or copper plates, having dimensions of $1 \times 1 \text{ cm}^2$. The moving coil meter can be a small VU meter

with 1-kilo-ohm coil resistance and 0-10 digit reading.

After assembling the circuit, adjust the presets such that IC1 outputs around 6 volts. None of the LEDs (LED1 through LED3) glows in this position with the touch plates open.

Now gently touch the touch plates with your middle finger. Maintain the finger still allowing one minute to bond with the pads and keep your body relaxed. Adjust VR3 until the green LED (LED1) lights up and the meter shows full deflection. Adjust VR2 to get maximum deflection of the meter. This indicates normal resistance of the skin, provided the body is fully relaxed.

If you are stressed or have ill feeling, skin resistance decreases and the blue LED lights up followed by the red LED along with a deflection of the meter towards the lower side. In short, the red LED and zero meter reading indicate you are stressed, and the green LED and high meter reading indicate you are relaxed. Practise some relaxation technique and observe how much your body is relaxed. ●

CLOCK TICK-TOCK SOUND GENERATOR & LED PENDULUM

■ ASHOK K. DOCTOR

Wooden-case, battery-operated wall clocks with pendulums are available

in the market. Some even have chimes. What is missing is the tick-tock sound of old mechanical pendulum clocks.

Fig. 1 shows the circuit of a tick-tock sound generator. It is built around timer IC

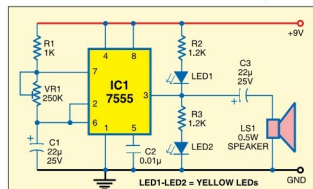


Fig. 1: Circuit of tick-tock sound generator



Fig. 2: Wooden clock

7555 (IC1) wired as an astable multivibrator. Resistor R1, preset VR1 and capacitor C1 generate frequencies that can be varied with the help of preset VR1. At output pin 3 of IC1, a small speaker is connected via capacitor C3 that produces the tick-tock sound. At pin 3 of IC1, two yellow LEDs are also connected on the clock face to flash in time with sound (see Fig. 2). Adjust preset VR1 such that ticking sound closely resembles the tick-tock of old pendulum clocks. Yellow LEDs (LED1 and

